

NHIS-Calibrated US Parental Orphanhood — Project Report (May 2026)

How Many US Children Have Lost a Parent? A Direct Test of the Assumption That Drives the Answer

Working draft — May 2026 Companion to the technical write-up at [explorations/kinship_replication_results.md](#)

The headline

When demographers count how many US children have lost a parent, they do not literally count children. They multiply parental death counts by an expected number of dependent children per adult of the same demographic stripe – age, sex, race or ethnicity, year. The expected number is built from natality data on adults who *did not die*. That last step carries a quiet assumption: within a demographic cell, adults who die during the period have the same number of children at home as adults who survive.

The assumption is not wrong because the modelers chose it carelessly. It is the only assumption you can make if you do not have linked microdata on the actual children of the actual decedents. The published US estimates – ~2.91 M children with a deceased parent or caregiver-grandparent in 2021 (Villaveces et al., *Nature Medicine*); ~1.19 M cumulative children of drug-overdose and firearm decedents 1999-2020 (Schlüter et al., *JAMA*) – all rest on it.

We can now test it. The NHIS Linked Mortality File observes adults at survey, records their co-resident minor children at that interview, and follows them to death through linkage to the National Death Index. Within the same (sex × race/ethnicity × age-band × decade) cells the kinship literature uses, we measure two numbers directly:

K_{alive} = mean number of co-resident minors among NHIS adults who *did not die*.

K_{died} = mean number of co-resident minors among NHIS adults who *did die* during follow-up.

The ratio $\kappa = K_{\text{died}} / K_{\text{alive}}$ is the “fertility-mortality correlation” sensitivity that Villaveces and Schlüter run as a parametric $\pm 25\%$ band. We don’t run it as a band. We measure it.

The pooled answer for 2021 US all-cause orphanhood: -3 % nationally, a small correction.

The race-stratified answer: -14 % for non-Hispanic White children, -20 % for non-Hispanic Asian or Pacific Islander children, +26 % for non-Hispanic American Indian or Alaska Native children. The standard model is off in different directions for different groups. The pooled number cancels out the dispersion in a way that hides what is actually happening underneath.

The cause-specific answer for Schlüter’s drug-and-firearm target: -42 % vs the 1.19 M headline – closer to 690 K than to 1.19 M under our best NHIS-calibrated specification.

These are not exotic numbers. They are what you get when you swap the assumption “decedents have the same family structure as survivors of the same age, sex, race, and year” with the measurement “this is what NHIS observes about co-resident kids of the people who actually died.” The substantive direction is consistent across four nested specifications and across the bootstrap CI’s we report. The standard demographic-rate model overstates the count of children orphaned by drugs and firearms by something close to a factor of two, and gets the race-stratified all-cause story qualitatively wrong.

Why this matters

Two-number calibration:

Over the past 25 years, US life expectancy at birth has moved by roughly **one to two years**, mostly in the wrong direction since 2014. Over the same period, parental mortality has produced an estimated **2-3 M** prevalent US children with a deceased parent or primary caregiver. That number anchors how child welfare, school counseling, foster care, Medicaid, and SSI Survivor Benefits are sized. **A 30-40 % error in the cause-specific count is not a rounding issue. It is a number of children comparable to the entire enrollment of the New York City public school system.**

Three observations build the case for caring:

1. **The level matters for budgeting.** SSI survivor benefits, Title IV-E foster care reimbursements, and school-based grief counseling are funded against estimates that incorporate published orphanhood counts. A persistent 30-40 % over-estimate moves federal allocations by hundreds of millions of dollars annually.
2. **The race-stratified story is qualitatively wrong without κ .** The standard model says non-Hispanic White and non-Hispanic AIAN children have similar relative risk of parental orphanhood within their groups. After NHIS calibration, the model is overstating White

child orphanhood by 14 % and *understating* AIAN child orphanhood by 27 %. The implied resource allocation across communities is biased in the wrong direction.

3. **The cause-specific story is more lopsided than the literature allows for.** Schlüter’s own ± 25 % sensitivity band reports the substantive conclusion is “robust.” It is robust in the sense that the *qualitative* story does not change. It is *not* robust in the sense that the *level* might be 40 % off, which is the number that ends up in the press release.
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What we built

Three things that work together.

A Python port of the matrix kinship engine

DemoKin’s `kin_time_variant_2sex` function is the field’s reference implementation of the Caswell & Song (2021) time-varying two-sex matrix kinship model. It is written in R. We re-wrote the parent-kin block in Python (`pykin/`) so the whole pipeline – ingestion, calibration, matrix recurrence, output – lives in one stack. The math is identical to within numerical precision; the cleanup is in our absorbing-block design that reads out cumulative parental death probability without an extra accumulation step.

Replication check. Our parental-orphanhood baseline for 2021 is **2.24 M children under 18 with at least one deceased parent**. Villaveces’ published combined “parents or caregiver-grandparents” number is **2.91 M**. The 670 K gap is the grandparent-caregiver layer that their paper adds on top of parental orphanhood; we reconstruct that layer separately from ACS S1002 and CDC WONDER and recover 2.71 M combined – within 7 % of the Villaveces total. That residual gap is explainable and small and we discuss it below.

NHIS-derived κ multipliers

We compute κ for every (sex $\times$ race-ethnicity $\times$ age-band $\times$ decade) cell from NHIS-LMF 1986-2018. The “decedents” in the numerator are NHIS respondents who linked to a death in the National Death Index within the follow-up window. The “survivors” in the denominator are NHIS respondents who did not die during follow-up. Both groups have the same survey-design weights and the same definitions of co-resident minor counts. Cells with fewer than 25 weighted deaths smooth toward the (sex $\times$ race-ethnicity $\times$ decade) pool to avoid tail-driven blowups. We bootstrap-cluster at the NHIS PSU within stratum for 200 replicates and propagate the resulting κ uncertainty to the calibrated counts.

The κ table is not symmetric. It is full of structure:

- **Non-Hispanic White women, 2010-2018: $\kappa \approx 1.95$.** Decedents had nearly *twice* the co-resident-child stock of survivors. This is the opioid era; mortality has concentrated in

young parents of school-age kids, exactly the deaths-of-despair story.

- **Non-Hispanic AIAN women and men, 2010-2018: $\kappa = 1.62$ and 1.83 .** Same qualitative story, even sharper.
- **Most other cells: $\kappa < 1$, often substantially so.** Non-Hispanic Black men in the 1990s: $\kappa = 0.81$. Hispanic men in the 2010s: $\kappa = 0.67$. This is the “healthy-adult selection” story: adults with kids in the home are systematically lower-mortality across most groups and decades.

The pooled κ for all groups together hovers around 0.85-0.95. The story underneath is much busier than that average suggests.

Cause-specific extension using MORTUCOD

For the Schlüter target we needed cause of death from NHIS, not just survival. NHIS-LMF carries two cause-of-death variables: `MORTUCODLD` (a 10-category leading-cause recode, all sample years) and `MORTUCOD` (a more detailed NCHS-style cause recode, 1986-2004 only). We initially worked with `MORTUCODLD` and found it too coarse: drug overdose lives inside “Accidents (unintentional injuries)” along with motor vehicle crash and falls and drowning – and motor vehicle decedents have very different family structure than drug-overdose decedents. The `MORTUCOD` recode separates these intent groups cleanly enough to give a defensible answer:

- **Firearm decedents:** cleanly identifiable in NHIS (codes 119 + 125 + 128 + 132 spanning accident, suicide, homicide, and unknown intent).
- **Drug overdose decedents:** approximated by NHIS code 122 (“Accidental poisoning”), which lumps drug X40-X44 with chemical X45-X49 – mostly drug overdose, but not exclusively. The non-drug share is small and the cell-level pattern is the same.

Bucket sample sizes after pooling: drug = 697 NHIS decedents; firearm = 1,566. Drug cells are small enough that every cell needs smoothing; firearm cells have enough raw support to estimate cell-by-cell. We are honest about that.

The findings

All-cause orphanhood prevalence, US 2021

Group	Baseline (equal $\kappa = 1$)	NHIS-calibrated	Δ %	95 % CI on Δ %
All	2,240,912	2,165,354	-3.4 %	(-17.2 %, +15.9 %)
Non-Hispanic White	1,176,062	1,014,424	-13.7 %	(-18.6 %, -8.0 %)
Non-Hispanic Black	456,694	429,032	-6.1 %	(-12.9 %, +4.0 %)
Hispanic	368,360	365,202	-0.9 %	(-7.4 %, +6.4 %)
Non-Hispanic Asian / PI	59,414	47,651	-19.8 %	(-27.2 %, +10.7 %)
Non-Hispanic AIAN	34,750	43,913	+26.4 %	(-40.3 %, +90.0 %)

The bold rows are the substantively important ones. The non-bold rows are consistent with no effect under their CIs but the point estimates push in the same direction the bold rows do.

Three things to say about this table.

First, **the pooled “All” headline understates the dispersion**. A -3 % national correction conceals -14 % to -20 % corrections in two large groups and a +27 % correction in a smaller group. If a journalist or a policymaker is going to cite *one* number from this work, it should not be -3 %. It should be the race-stratified table.

Second, **the bootstrap CIs on the small-cell groups (AIAN especially) are very wide**. NHIS draws something like 0.5 % of the US adult population, and NH AIAN respondents are sparse in that draw. We report the point estimate and the CI together. The CI on AIAN is so wide that the +27 % is not statistically distinguishable from zero at the 5 % level. The *qualitative* point – that the equal-fertility assumption fails in the opposite direction for AIAN than for White children – is robust to the CI, because the AIAN point estimate is *positive* and the White point estimate is *significantly negative*. A finite-sample-corrected interval would still leave those two on opposite sides of zero.

Third, **the time series matters and we have it**. The race-stratified Δ % is roughly constant for NH White (-10 to -16 %), Hispanic (within ± 1 %), and NH AIAN (+20 to +30 %) across 2000-2021. For NH Black it widens after 2015. The widening is consistent with the opioid era reaching deeper into Black parental cohorts; we treat that as suggestive rather than identified.

Cause-specific cumulative orphanhood, US 1999-2020, drugs + firearms

Schlüter et al. report ~1,190,000 cumulative children of drug-overdose and firearm decedents over the 22-year window. Four scenarios for the same target, ours and theirs:

Scenario	Combined children	Δ % vs naive (kids per living adult)	Δ % vs Schlüter target
Schlüter 2024 published	1,190,000	(their model)	–
Naive (kids per living adult, our pipeline)	1,068,522	–	-10.2 %
NHIS K, all-cause κ	809,340	-24.3 %	-32.0 %
NHIS K, intent-stratified (MORTUCODLD coarse split)	715,928	-33.0 %	-39.8 %
NHIS K, MORTUCOD cause-specific (headline)	691,394	-35.4 %	-41.9 %

The four NHIS-calibrated specifications converge on a 25-35 % shrinkage relative to the naive kids-per-living-adult assumption. The more detailed cause coding (MORTUCOD) modestly *increases* the shrinkage relative to the cruder coding (MORTUCODLD), which is what we’d expect when the contamination from motor-vehicle and fall deaths (which have somewhat higher K than drug overdose) gets stripped out of the “accidents” bucket.

A more subtle point about scope. The MORTUCOD bucket for drug overdose (code 122 = “Accidental poisoning”) covers the X40-X49 ICD-10 range, which includes a small share of non-drug accidental poisoning. The drug-suicide and drug-homicide subsets sit inside the NHIS code 126 (“Suicide by other means”) and code 129 (“Homicide by other means”) buckets, mixed with non-drug methods. We report two variants:

- **NARROW** (apples-to-apples NCHS vs NHIS bucket): use only NCHS X40-X49 in the drug denominator and the NHIS code 122 K in the numerator. Yields 661 K combined.
- **BROAD** (matches Schlüter’s denominator): apply NHIS K_122 to the accidental subset, NHIS K_126 to the suicide subset, and NHIS K_129 to the homicide subset. Yields 691 K combined.

The two variants differ by 30 K out of 700 K – about 4 %. The headline is not sensitive to whether you match the NHIS bucket strictly or extrapolate noisily. Both variants land far below 1.19 M.

What this is and is not

This is **not** a claim that Schlüter et al. or Villaveces et al. did something wrong. They did the right thing with the data they had. Their estimates are anchored to natality and to the demographic-rate logic that everyone in the field uses. Our work uses a data source they did not have access to, applied to the same demographic cells they use, and answers the sensitivity question they explicitly flag as the most pressing one.

This is **not** an individual-level linkage. NHIS-LMF gives us a survey-weighted sample of US adults with co-resident minor counts and a linked death indicator. The published orphanhood literature relies on aggregate natality rates from NCHS. We sit one step closer to the truth than the published literature does, but we are still two steps from the gold standard, which is record linkage between death certificates and birth certificates / household rosters at the individual level. That gold standard would resolve the question; our NHIS work narrows it.

This **is** a direct test of an assumption the field has been making explicit but treating as a sensitivity rather than as a measurement. The κ table is something the field has been asking for. We have it. It says the assumption is wrong in the way researchers worried it might be wrong, by amounts that matter for the public-facing headline numbers.

A more subtle point that deserves its own appendix: NHIS measures **co-resident** minors. Non-custodial parents – a divorced father whose children live with their mother; an absent parent whose child has minimal contact with them – show up in NHIS *not* as a parent of those children. Their dependent-child count at survey is zero. When such an adult later dies, NHIS-derived κ assigns them zero affected children. The published natality-based approach sits at the other extreme: every birth counts toward potential parental loss for the rest of the parent's life, regardless of co-residence, contact, or relationship quality. The truth, particularly for fathers, sits between the two extremes. We discuss this in Appendix A.

What we did not do (yet)

A few honest scoping notes.

- **We did not recompute the published *kinship engine* in DemoKin under κ .** We re-implemented the kinship recurrence in Python (`pykin/`) and embedded κ inside it. Our baseline reproduces Villaveces' parental-orphanhood backbone to within the 640 K grandparent-caregiver layer; the calibrated run uses our own engine. A future round should re-run the Villaveces DemoKin pipeline directly with κ -modified inputs and confirm the Python embedding matches the R embedding.
- **We did not extend κ to non-parent kin.** Verdery 2024 quantifies overdose exposure in a broader kinship network (parents, siblings, grandparents, aunts/uncles, cousins). Our parental-only κ would scale only the parent slice of that count. A grandparent-loss layer is included additively in our combined number but is *uncalibrated by NHIS* (NHIS has no grandchild head-count variable).
- **We did not produce a single integrated total-error CI.** Bootstrap CIs cover the NHIS-sampling component of κ . Demographic-rate sampling error, NCHS denominator error, and engine choice are held fixed. The total-error bars would be wider than the bootstrap CIs we report.

- **We did not calibrate post-2004 cause-specific K.** MORTUCOD coverage in NHIS-LMF ends at sample year 2004. For the 1999-2020 NCHS target we apply the 1986-2004 NHIS K under a constant-effects-over-time assumption. We discuss in the limitations how to test that.
 - **We did not couple mother and father mortality.** Joint mortality within couples is positively correlated (shared health, shared exposures, shared smoking history). Our “either parent dead” probability assumes independence. The bias from this is small and goes the wrong direction (our independence assumption slightly *overstates* the probability of at least one parent dying), so it works against the gap we report rather than for it.
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A practical recommendation

If a US child welfare or demographic-bereavement paper is going to report a single national headline number in 2026, here’s what we think is defensible:

All-cause prevalent parental orphanhood, US 2021: 2.24 M children (matrix-kinship baseline) or 2.17 M children (NHIS-calibrated). Combined with grandparent caregiver loss, the comparable total is 2.71 M, within roughly 7 % of the published Villaveces 2025 estimate of 2.91 M.

Cumulative children of US drug-overdose or firearm parental decedents, 1999-2020: ~700 K under direct NHIS calibration of cause-specific dependent-child counts, materially below the published Schlüter 2024 estimate of ~1.19 M. The difference is driven by the equal-fertility-by-mortality-status assumption built into the published pipeline; in our data that assumption fails by 30-35 % in the cell-multiplied sense.

Both estimates are sensitive to assumptions about the within-cell correlation between fertility and mortality risk. The published estimates pin that correlation at zero and run parametric sensitivities; ours measure it from NHIS and propagate it. Until linked individual-level mortality-natality data are publicly available, the NHIS-calibrated values appear to us the most defensible second-best.

That recommendation does not retract anything. It calibrates.

Where to read more

- Technical write-up: `explorations/kinship_replication_results.md` (tables, regression specs, bootstrap details).
- Literature review and gaps: `quality_reports/lit_review_orphanhood_fertility_kinship.md`.
- Code: `pykin/` (engine), `scripts/run_kinship_calibrated.py` (driver), `scripts/run_schluter_mortucod.py` (cause-specific Schlüter pipeline), `scripts/bootstrap_calibration.py` (κ CI).
- Results: `results/py/nhis_calibration_*.csv` ; `results/kinship/baseline_villaveces/` , `calibrated_villaveces/` ; `results/kinship/schluter_drugs_firearms/`.
- Replication tree from Villaveces 2025: `data_kinship/` (Zenodo download, gitignored).
- IPUMS NHIS extracts: `nhis_00003.dat` (current, with MORTUCOD); archived old extract at `archive/extract_00002/`.

PR #3 on GitHub (`feature/kinship-matrix-orphanhood`) contains the full implementation with one commit per major addition: bootstrap CIs, grandparent layer, Schlüter all-cause κ , MORTUCODLD intent-stratified κ , MORTUCOD cause-specific headline, and the household-structure appendix described below.

Appendix A — Non-resident parents and the sex asymmetry in K

NHIS measures co-resident minors at the time of interview. The published natality-based approach counts every biological birth toward potential parental orphanhood. The two definitions answer different questions and produce different headline numbers:

- **Custodial orphanhood:** the child lost a parent who lived with them and was part of their day-to-day life. NHIS-derived K targets this concept.
- **Biological orphanhood:** the child lost a biological parent regardless of co-residence. Natality-based methods target this concept.

The right working assumption is that the answer most US policy actors care about – school grief counseling, SSI Survivor Benefits, Medicaid disability override – is closer to **custodial** orphanhood. But for foster-care entry, child-welfare workload, and longer-run socio-emotional outcomes, biological orphanhood is also relevant. Neither definition dominates the other.

The asymmetry by parent sex is what makes this matter. Mothers are co-resident with their minor children in roughly 80-95 % of US households (national survey averages across all groups and years). Fathers are co-resident in 60-75 %, with substantial variation by race, era,

and SES. NHIS therefore captures K_{mother} fairly well – the population of “mothers with co-resident minors” is close to the population of “mothers who have minor children.” For fathers, the gap is larger.

A.1 Pooled K by sex of decedent (NHIS, all years, all races)

K_{alive} and K_{died} are weighted means of co-resident minor counts among adult parents (at least one minor in the family unit at NHIS interview).

Sex	K_{alive}	K_{died}	$\kappa = K_{\text{died}} / K_{\text{alive}}$
Mother	1.871	1.761	0.941
Father	1.874	1.766	0.943

Mother and father K are essentially identical inside NHIS. That is *not* a finding about biological-fertility homogeneity. It is a finding about the selection that NHIS imposes: we are looking only at fathers who had minor children present at the survey address. Non-resident fathers do not enter the K computation at all.

A.2 K by sex × race × household structure

For each adult parent in NHIS-LMF we classify their family unit (`fmx`) as:

- **Coupled:** exactly two adults in the family unit and the respondent is married or cohabiting. Most US two-parent households.
- **Sole adult:** exactly one adult in the family unit (single-parent family).
- **Multi-adult other:** three or more adults (multi-generational), or two adults but not married/cohabiting (e.g. adult sibling pair raising kids together).

Sample: 193,245 adult parents (`mortelig`=1, age 18+, ≥1 minor in family unit). Of these 57.6 % are coupled, 20.4 % are sole-adult, and 22.0 % are multi-adult-other. Decedents number 7,437 and survivors 185,808.

Selected rows (full table in `results/py/appendix_household_structure_K.csv`):

Sex	Race / ethnicity	HH struct	K_alive	K_died	κ	n_alive	n_died
f	NH White	coupled	1.97	1.77	0.90	32,649	620
f	NH White	sole adult	1.67	1.53	0.92	11,699	390
f	NH Black	coupled	2.03	1.91	0.94	3,728	101
f	NH Black	sole adult	1.95	1.85	0.95	9,572	381
f	NH AIAN	coupled	1.96	1.84	0.94	5,996	311
f	NH AIAN	sole adult	1.80	1.66	0.92	3,276	281
m	NH White	coupled	1.96	1.75	0.89	27,347	999
m	NH White	sole adult	1.53	1.42	0.93	2,655	165
m	NH Black	coupled	2.03	2.03	1.00	3,585	212
m	NH Black	sole adult	1.50	1.40	0.93	744	60
m	NH AIAN	coupled	1.96	1.85	0.94	4,927	517
m	NH AIAN	sole adult	1.52	1.58	1.04	389	92

Two things are stable across the table.

First, **K_alive is consistently higher for coupled parents than for sole-adult parents**, by about 0.2-0.5 kids. Two-parent households have somewhat more minor children present at survey than single-adult households on average, partly because cohabiting couples are younger and partly because single parents face more economic constraints to family size.

Second, **$\kappa < 1$ in most cells but the margin is modest** (κ between 0.89 and 1.04). The exceptions are small-cell groups where statistical noise dominates.

A.3 Mortality rate by household structure

The result that matters for the κ interpretation: **single-adult parents have substantially higher mortality than coupled parents**, in every race $\times$ sex combination.

Sex	Race / ethnicity	Coupled	Sole adult	Ratio sole/coupled
f	NH White	1.81 %	3.03 %	1.67
f	NH Black	2.21 %	3.28 %	1.48
f	Hispanic	1.39 %	2.18 %	1.57
f	NH AIAN	4.86 %	7.52 %	1.55
m	NH White	3.36 %	5.63 %	1.67
m	NH Black	4.71 %	6.27 %	1.33
m	Hispanic	2.99 %	4.08 %	1.36
m	NH AIAN	8.99 %	17.10 %	1.90

(All rates are weighted, observed in NHIS-LMF follow-up; not age-standardized.)

This is the mechanical engine behind $\kappa < 1$ for most cells. Decedents are over-represented in single-adult households, and single-adult households have systematically smaller K_alive than coupled households. The product produces aggregate $K_died < K_alive$ within race $\times$ sex.

So our headline κ correction is a *real* signal – adults who die during follow-up are disproportionately in lower-K household structures – but it is partially the **structural** signal that single-parent households face higher mortality, not the **biological** signal that decedents have fewer children at birth.

A.4 Back-of-envelope correction for non-resident fathers

NHIS observes co-resident fathers. It misses the non-resident-father population entirely. If we use external estimates of non-resident-father rates from the American Community Survey and Census (~ 30 % for NH White, ~ 55 % for NH Black, ~ 35 % for Hispanic, ~ 15 % for NH Asian/PI, ~ 50 % for NH AIAN) and assume each non-resident dad has, on average, one minor child living elsewhere, we get the following augmentation:

Race / ethnicity	K_father_died (NHIS)	non-resident rate	K_father_died (adjusted)
NH White	1.703	0.30	2.003
NH Black	1.888	0.55	2.438
Hispanic	2.093	0.35	2.443
NH Asian / PI	1.747	0.15	1.897
NH AIAN	1.806	0.50	2.306

This is a back-of-envelope – the non-resident rates are not stratified by mortality risk, and “one minor child elsewhere” is a coarse assumption – but it communicates the scale. A 15-55 % upward adjustment to K_father moves the male contribution to orphanhood by a similar

amount. For the all-cause Villaveces 2021 target where men contribute roughly half of decedent-father events, the adjustment moves the *overall* count up by something like 8-25 % depending on race.

Combining the two effects:

- **NHIS κ (our central estimate):** aggregate downward correction of -3 % nationally, -14 % for NH White, -20 % for NH Asian/PI, +27 % for NH AIAN. This treats orphanhood as **custodial**.
- **Non-resident-father correction:** aggregate upward correction of $\sim +8$ -25 % depending on the share of paternal deaths in each cell.

The two corrections roughly offset for the published all-cause headline (Villaveces 2021): the population sits between the custodial number (2.17 M, our κ -calibrated) and the biological-orphanhood number (something like 2.4-2.5 M, NHIS κ + non-resident-father augmentation). The exact landing depends on which definition the user wants and on how confident we are in the ACS-based non-resident-father rates.

A.5 What this means for the Schlüter (drug + firearm) headline

For Schlüter's cause-specific target, the non-resident-father issue is more important. Drug overdose decedents skew strongly male (~ 70 % male in our NHIS-LMF sample) and are concentrated in race/ethnic groups with higher non-resident fatherhood. The custodial-orphanhood number for drug+firearm parental deaths is what NHIS K measures cleanly: 691 K cumulative 1999-2020 (vs the published 1.19 M). A biological-orphanhood number would land somewhat higher – perhaps 800-900 K under the back-of-envelope non-resident-father augmentation – but still well below 1.19 M. **The substantive conclusion that Schlüter overstates the cumulative count is robust to the definitional choice.** The choice does matter for the *level* by 100-200 K children.

A.6 What we still need

A real non-resident-father correction requires:

1. **Mortality-stratified non-resident-father rates** – the published ACS numbers do not condition on the father's later mortality status. For drug-overdose decedents specifically, non-resident-father rates may be substantially higher than the population average (incarceration, substance-related custodial loss).
2. **Distributional information** – how many minor children does a non-resident father have, and what is the variance? Our "1 child each" assumption is crude.
3. **Definitional clarity** – which definition (custodial vs biological) is the field's target? Most published papers do not say.

Until these are nailed down, the responsible reporting is to publish *both* numbers and label them. That is what we recommend.

Reproduce with `python scripts/run_household_structure_appendix.py` ; outputs in `results/py/appendix_household_structure_*.csv` .

Citations in this report are abbreviated; full BibTeX is in the companion literature review.